

C++ Assignment 2

Question 1 — Inventory Management System

Part A — Product class with Array of Objects

```
#include <iostream>
#include <string>
using namespace std;

class Product
{
private:
    int productId;
    string name;
    double price;
    int quantity;

public:

    void acceptDetails()
    {
        cout << "Enter Product ID: ";
        cin >> productId;

        cout << "Enter Product Name: ";
        cin >> name;

        cout << "Enter Price: ";
        cin >> price;

        cout << "Enter Quantity: ";
        cin >> quantity;
    }

    void displayDetails() const
    {
        cout << productId << "\t"
            << name << "\t"
            << price << "\t"
            << quantity << "\t"
```

```

        << totalValue() << endl;
    }

double totalValue() const
{
    return price * quantity;
}

// Check low stock
bool isLowStock(int threshold) const
{
    return quantity < threshold;
}
};

int main()
{
    Product products[5];

    for (int i = 0; i < 5; i++)
    {
        cout << "\nEnter details of Product " << i + 1 << endl;
        products[i].acceptDetails();
    }

    cout << "\n===== INVENTORY REPORT =====\n";

    cout << "ID\tName\tPrice\tQty\tTotal Value\n";

    for (int i = 0; i < 5; i++)
    {
        products[i].displayDetails();
    }

    int highest = 0;

    for (int i = 1; i < 5; i++)
    {
        if (products[i].totalValue() > products[highest].totalValue())

```

```

        {
            highest = i;
        }
    }

    cout << "\nHighest Value Product:\n";
    products[highest].displayDetails();

    int threshold;

    cout << "\nEnter Low Stock Threshold: ";
    cin >> threshold;

    cout << "\nLow Stock Products:\n";

    for (int i = 0; i < 5; i++)
    {
        if (products[i].isLowStock(threshold))
        {
            products[i].displayDetails();
        }
    }

    return 0;
}

```

```

===== INVENTORY REPORT =====
ID      Name      Price  Qty    Total Value
1001    Widget-A250    120    30000
1002    Bolt-B    15.5    8      124
1003    Nut-C     20      50     1000
1004    Screw-D  10      5       50
1005    Gear-E    500     20    10000

Highest Value Product:
1001    Widget-A250    120    30000

Enter Low Stock Threshold: 10
|

Low Stock Products:
1002    Bolt-B    15.5    8      124
1004    Screw-D  10      5       50

```

Part B — Function Overloading

```
#include <iostream>
using namespace std;

double reorderCost(int qty, double unitPrice)
{
    return qty * unitPrice;
}

double reorderCost(double qty, double unitPrice)
{
    return qty * unitPrice;
}

double reorderCost(int qty, double unitPrice, double taxRate)
{
    double cost = qty * unitPrice;
    double tax = cost * taxRate / 100;

    return cost + tax;
}

double applyDiscount(double price, double discountPercent = 10.0)
{
    double discount = price * discountPercent / 100;

    return price - discount;
}

int main()
{
    double cost1 = reorderCost(10, 50.0);

    double cost2 = reorderCost(5.5, 100.0);

    double cost3 = reorderCost(10, 50.0, 18.0);

    double price1 = applyDiscount(1000, 20);

    double price2 = applyDiscount(1000);

    cout << "Integer Quantity Cost: " << cost1 << endl;
```

```

cout << "Fractional Quantity Cost: " << cost2 << endl;

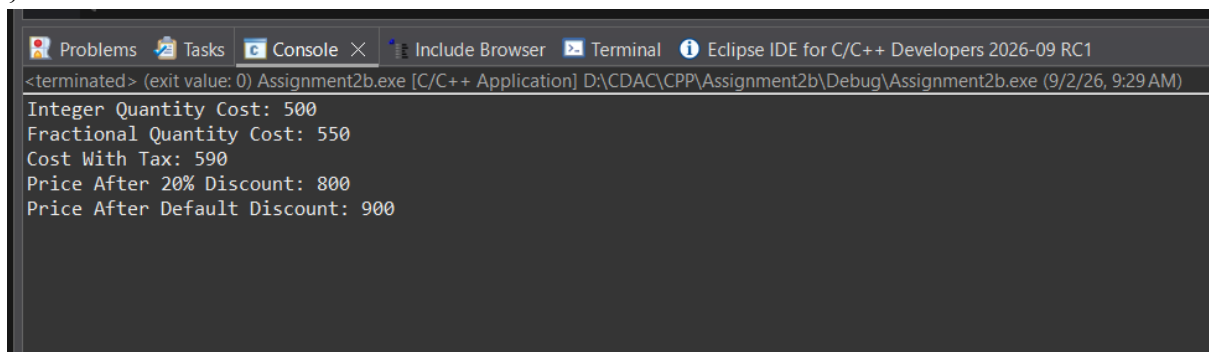
cout << "Cost With Tax: " << cost3 << endl;

cout << "Price After 20% Discount: " << price1 << endl;

cout << "Price After Default Discount: " << price2 << endl;

return 0;
}

```



The screenshot shows the Eclipse IDE interface with the 'Console' tab selected. The output of the program is as follows:

```

<terminated> (exit value: 0) Assignment2b.exe [C/C++ Application] D:\CDAC\CPP\Assignment2b\Debug\Assignment2b.exe (9/2/26, 9:29 AM)
Integer Quantity Cost: 500
Fractional Quantity Cost: 550
Cost With Tax: 590
Price After 20% Discount: 800
Price After Default Discount: 900

```

Question 2 — Hospital Patient Registry

```

#include <iostream>
#include <string>
using namespace std;

class Patient
{
private:
    int patientId;
    string name;
    int age;
    string ward;
    const string bloodGroup;

public:
    Patient()
        : patientId(0), name("Unknown"), age(0),
          ward("General"), bloodGroup("O+")
    {
        cout << "[Constructor] Default patient registered." << endl;
    }
}

```

```
}
```

```
Patient(int id, const string& name)
    : patientId(id), name(name), age(0),
      ward("Emergency"), bloodGroup("O+")
{
    cout << "[Constructor] Emergency: " << name << endl;
}
```

```
Patient(int id, const string& name, int age,
        const string& ward, const string& bg)
    : patientId(id), name(name), age(age),
      ward(ward), bloodGroup(bg)
{
    cout << "[Constructor] Full admission: " << name << endl;
}
```

```
~Patient()
{
    cout << "[Destructor] Patient " << name
          << " discharged." << endl;
}
```

```
void displayRecord() const
{
    cout << "\nPatient Record:" << endl;
    cout << " ID      : " << patientId << endl;
    cout << " Name    : " << name << endl;
    cout << " Age     : " << age << endl;
    cout << " Ward     : " << ward << endl;
    cout << " Blood Grp : " << bloodGroup << endl;
}
```

```
void transferWard(const string& newWard)
{
    cout << "\nWard Transfer: " << name
          << " -> " << newWard << endl;

    ward = newWard;
}
};
```

```
int main()
```

```

{

    Patient p1(1001, "Meera Joshi", 34,
               "Cardiology", "B+");

    Patient p2(1002, "Raj Patel");

    Patient p3;

    Patient* patients = new Patient[4];

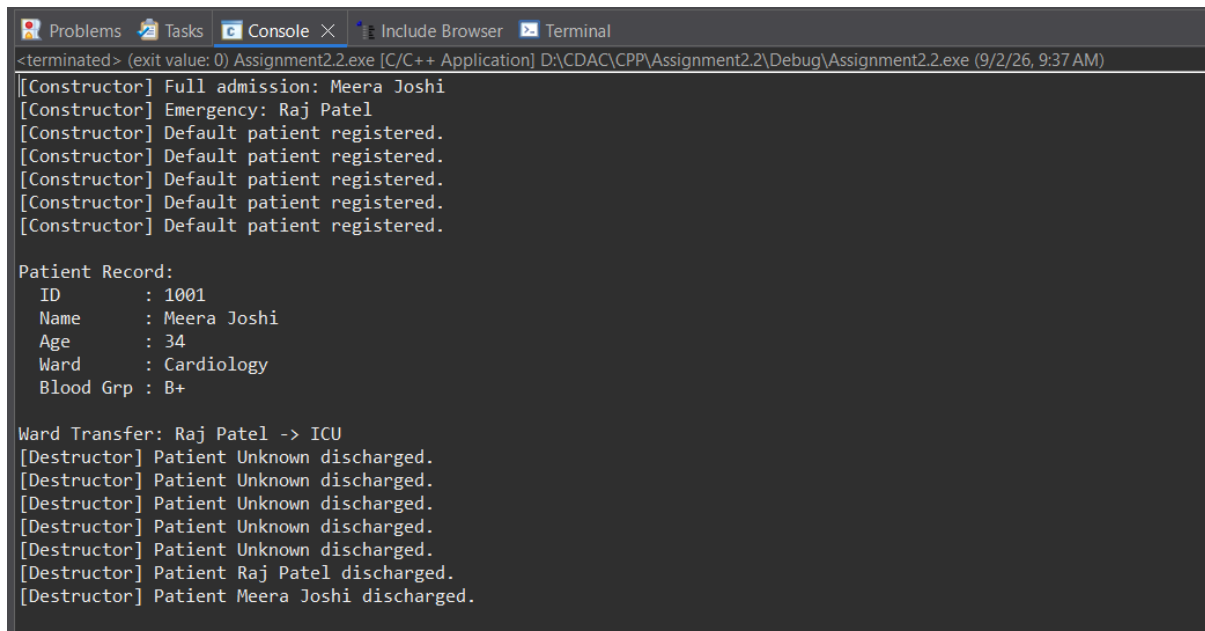
    p1.displayRecord();

    p2.transferWard("ICU");

    delete[] patients;

    return 0;
}

```



The screenshot shows a C++ IDE with a console window. The console output is as follows:

```

<terminated> (exit value: 0) Assignment2.2.exe [C/C++ Application] D:\CDAC\CPP\Assignment2.2\Debug\Assignment2.2.exe (9/2/26, 9:37 AM)

[Constructor] Full admission: Meera Joshi
[Constructor] Emergency: Raj Patel
[Constructor] Default patient registered.
[Constructor] Default patient registered.
[Constructor] Default patient registered.
[Constructor] Default patient registered.
[Constructor] Default patient registered.

Patient Record:
  ID      : 1001
  Name    : Meera Joshi
  Age     : 34
  Ward    : Cardiology
  Blood Grp : B+

Ward Transfer: Raj Patel -> ICU
[Destructor] Patient Unknown discharged.
[Destructor] Patient Unknown discharged.
[Destructor] Patient Unknown discharged.
[Destructor] Patient Unknown discharged.
[Destructor] Patient Unknown discharged.
[Destructor] Patient Raj Patel discharged.
[Destructor] Patient Meera Joshi discharged.

```

Question 3 — Game Engine Entity System

Part A — this Pointer and Method Chaining

```

#include <iostream>
#include <string>

```

```
using namespace std;
```

```
class Entity
```

```
{
```

```
private:
```

```
    string name;
```

```
    int health;
```

```
    int level;
```

```
    string type;
```

```
public:
```

```
    Entity& setName(const string& name)
```

```
    {
```

```
        this->name = name;
```

```
        return *this;
```

```
    }
```

```
    Entity& setHealth(int health)
```

```
    {
```

```
        this->health = health;
```

```
        return *this;
```

```
    }
```

```
    Entity& setLevel(int level)
```

```
    {
```

```
        this->level = level;
```

```
        return *this;
```

```
    }
```

```
    Entity& setType(const string& type)
```

```
    {
```

```
        this->type = type;
```

```
        return *this;
```

```
    }
```

```
    string getName() const
```

```
    {
```

```
        return name;
```

```
    }
```

```
    int getHealth() const
```

```
    {
```

```
        return health;
```

```
    }
```

```

int getLevel() const
{
    return level;
}

string getType() const
{
    return type;
}

void displayInfo() const
{
    cout << "\nEntity Information:" << endl;
    cout << "Name   : " << name << endl;
    cout << "Health : " << health << endl;
    cout << "Level  : " << level << endl;
    cout << "Type   : " << type << endl;
}
};

```

```

int main()
{
    Entity player, enemy, item;

    player.setName("Aragorn")
        .setHealth(100)
        .setLevel(10)
        .setType("Player");

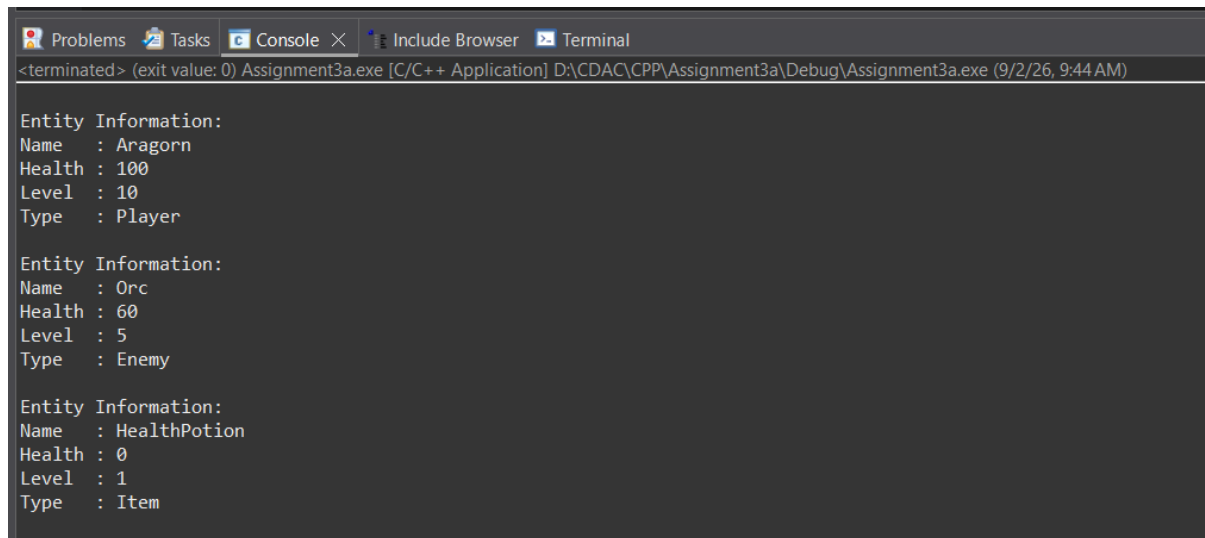
    enemy.setName("Orc")
        .setHealth(60)
        .setLevel(5)
        .setType("Enemy");

    item.setName("HealthPotion")
        .setHealth(0)
        .setLevel(1)
        .setType("Item");

    player.displayInfo();
    enemy.displayInfo();
    item.displayInfo();
}

```

```
    return 0;
}
```



```
<terminated> (exit value: 0) Assignment3a.exe [C/C++ Application] D:\CDAC\CPP\Assignment3a\Debug\Assignment3a.exe (9/2/26, 9:44 AM)

Entity Information:
Name   : Aragorn
Health : 100
Level  : 10
Type   : Player

Entity Information:
Name   : Orc
Health : 60
Level  : 5
Type   : Enemy

Entity Information:
Name   : HealthPotion
Health : 0
Level  : 1
Type   : Item
```

Part B — Namespaces

```
#include <iostream>
using namespace std;
namespace Physics
{
    double clamp(double val, double min, double max)
    {
        if (val < min)
            return min;
        else if (val > max)
            return max;
        else
            return val;
    }

    double lerp(double a, double b, double t)
    {
        return a + (b - a) * t;
    }
}
namespace GameMath
{
    int clamp(int val, int min, int max)
    {
```

```

        if (val < min)
            return min;
        else if (val > max)
            return max;
        else
            return val;
    }
    double lerp(double a, double b, double t)
    {
        return a + (b - a) * t;
    }
}

int main()
{
    double velocity = Physics::clamp(120.5, 0.0, 100.0);
    int health = GameMath::clamp(120, 0, 100);

    cout << "Clamped velocity: " << velocity << endl;
    cout << "Clamped health: " << health << endl;

    double physicsResult = Physics::lerp(10.0, 20.0, 0.5);
    double animationResult = GameMath::lerp(0.0, 100.0, 0.25);

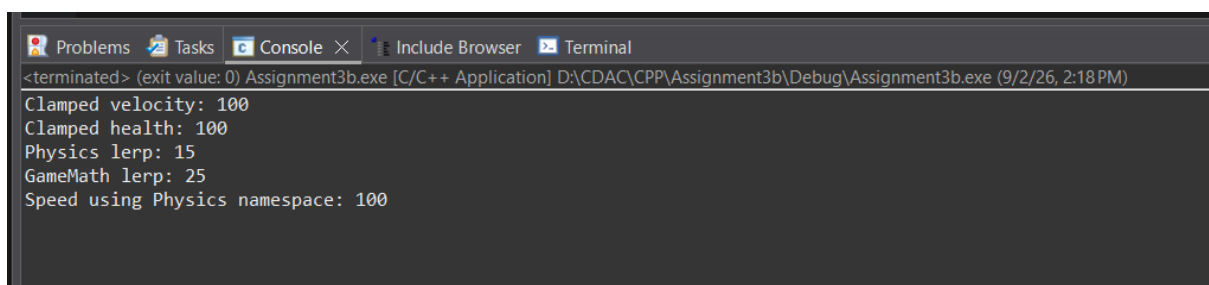
    cout << "Physics lerp: " << physicsResult << endl;
    cout << "GameMath lerp: " << animationResult << endl;
    {
        using namespace Physics;

        double speed = clamp(150.0, 0.0, 100.0);

        cout << "Speed using Physics namespace: " << speed << endl;
    }

    return 0;
}

```



The screenshot shows a terminal window with the following output:

```

<terminated> (exit value: 0) Assignment3b.exe [C/C++ Application] D:\CDAC\CPP\Assignment3b\Debug\Assignment3b.exe (9/2/26, 2:18 PM)
Clamped velocity: 100
Clamped health: 100
Physics lerp: 15
GameMath lerp: 25
Speed using Physics namespace: 100

```

Part C — Dynamic 2D Game Map

```
#include <iostream>
#include <cstdlib>
#include <ctime>

using namespace std;

int main()
{
    int R, C;

    // Read rows and columns
    cout << "Enter number of rows: ";
    cin >> R;

    cout << "Enter number of columns: ";
    cin >> C;

    // Allocate outer array
    int** map = new int*[R];

    // Allocate each row
    for (int i = 0; i < R; i++)
    {
        map[i] = new int[C];
    }

    // Generate random tile values
    srand(time(0));

    for (int i = 0; i < R; i++)
    {
        for (int j = 0; j < C; j++)
        {
            map[i][j] = rand() % 5;
        }
    }

    // Display map
    cout << "\n===== GAME MAP (" << R << " x " << C << ") =====" << endl;
```

```

for (int i = 0; i < R; i++)
{
    for (int j = 0; j < C; j++)
    {
        cout << map[i][j] << " ";
    }
    cout << endl;
}

// Legend
cout << "\nLegend: 0=Grass 1=Water 2=Mountain 3=Forest 4=Dungeon"
    << endl;

// Count each tile
int count[5] = {0};

for (int i = 0; i < R; i++)
{
    for (int j = 0; j < C; j++)
    {
        count[map[i][j]]++;
    }
}

// Display tile count
cout << "\nTile Count:" << endl;
cout << "Grass   : " << count[0] << endl;
cout << "Water   : " << count[1] << endl;
cout << "Mountain : " << count[2] << endl;
cout << "Forest   : " << count[3] << endl;
cout << "Dungeon  : " << count[4] << endl;

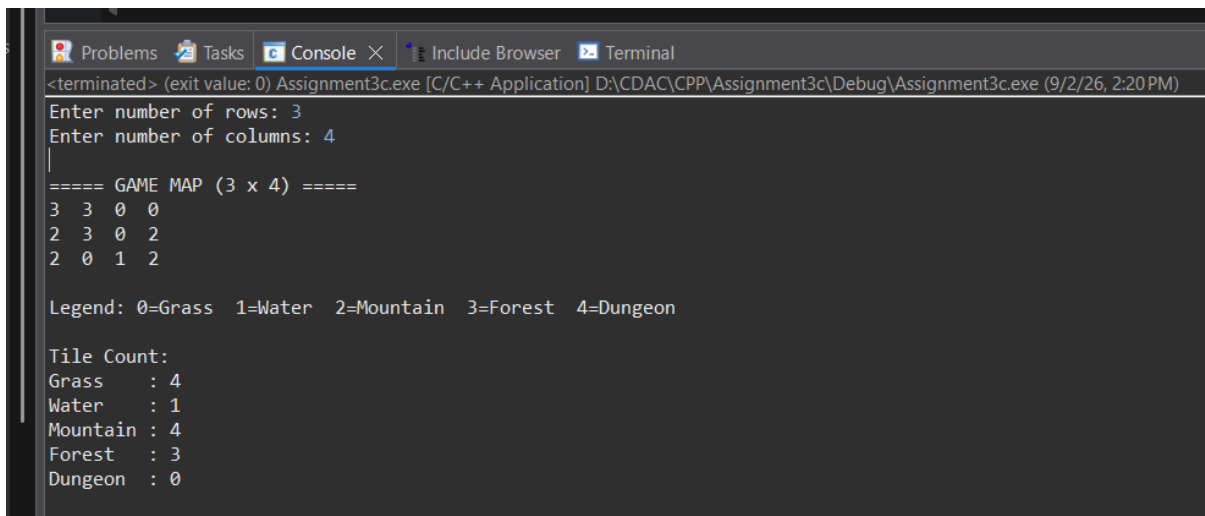
// Deallocate memory
for (int i = 0; i < R; i++)
{
    delete[] map[i];
}

delete[] map;

return 0;

```

}

A screenshot of a C++ application's console output. The window title bar shows 'Problems', 'Tasks', 'Console', 'Include Browser', and 'Terminal'. The console text shows the program has terminated with exit value 0. It prompts for the number of rows (3) and columns (4). It then displays a 3x4 game map with values 0-4. A legend identifies 0 as Grass, 1 as Water, 2 as Mountain, 3 as Forest, and 4 as Dungeon. Finally, it shows a tile count: Grass (4), Water (1), Mountain (4), Forest (3), and Dungeon (0).

```
<terminated> (exit value: 0) Assignment3c.exe [C/C++ Application] D:\CDAC\CPP\Assignment3c\Debug\Assignment3c.exe (9/2/26, 2:20 PM)
Enter number of rows: 3
Enter number of columns: 4

===== GAME MAP (3 x 4) =====
3 3 0 0
2 3 0 2
2 0 1 2

Legend: 0=Grass 1=Water 2=Mountain 3=Forest 4=Dungeon

Tile Count:
Grass : 4
Water : 1
Mountain : 4
Forest : 3
Dungeon : 0
```

Scope Resolution Bonus (Optional)

```
#include <iostream>
```

```
#include <string>
```

```
using namespace std;
```

```
// Global level
```

```
int level = 1;
```

```
// Nested namespace
```

```
namespace Engine
```

```
{
```

```
    namespace Audio
```

```
    {
```

```
        void playSound(string name)
```

```
        {
```

```
            cout << "Playing: " << name << endl;
```

```
        }
```

```
    }
```

```
}
```

```
int main()
```

```
{
```

```
    // Local level
```

```
    int level = 10;
```

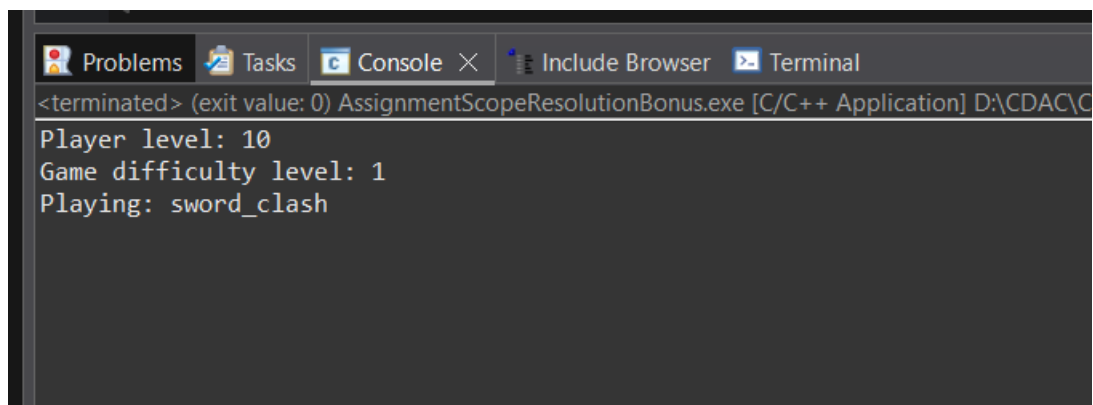
```
    // Print local level
```

```
    cout << "Player level: " << level << endl;
```

```
// Print global level using scope resolution operator
cout << "Game difficulty level: " << ::level << endl;

// Call nested namespace function
Engine::Audio::playSound("sword_clash");

return 0;
}
```



The screenshot shows a C++ IDE's console window. The title bar includes tabs for 'Problems', 'Tasks', 'Console', 'Include Browser', and 'Terminal'. The 'Console' tab is active, displaying the output of the program. The output text is: 'Player level: 10', 'Game difficulty level: 1', and 'Playing: sword_clash'. Above the output, a status line indicates the program was terminated with an exit value of 0.

```
<terminated> (exit value: 0) AssignmentScopeResolutionBonus.exe [C/C++ Application] D:\CDAC\C
Player level: 10
Game difficulty level: 1
Playing: sword_clash
```